MILITARY STANDARD

MIL-STD-1320-109A (NAVY)

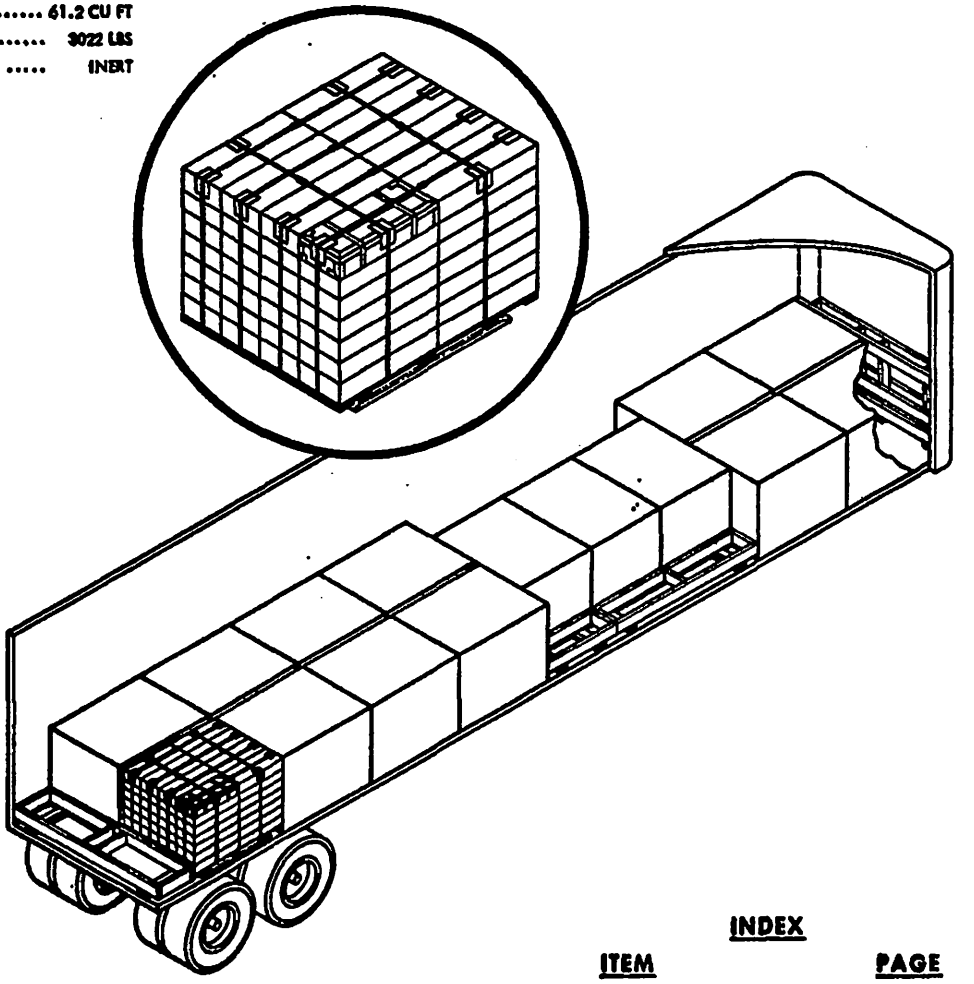
16 JUNE 1976

SUPERSEDING
MIL-STD-1320-109 (NAVY)
8 FEBRUARY 1975

TRUCKLOADING
BOMB, PRACTICE MK 76 MODS 4 AND 5
FLEET ISSUE UNIT LOAD

UNIT LOAD DATA

UNIT LOAD DRAWING	WR-54/217
LENGTH	55 3/4
WIDTH	44 1/2
HEIGHT	42 5/8
CUBE	61.2 CU FT
WEIGHT	3022 LBS
DOT HAZARD CLASSIFICATION	INERT



INDEX

ITEM	PAGE
GENERAL NOTES	2
FTL & LTL	2, 5 & 6
LOAD PATTERNS	3

NOTES:
1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES.
2. FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING, CONTAINERLOADING AND CARLOADING MILITARY STANDARDS, REFER TO INDEX TO STANDARDS, MIL-HDBK-236 (NAVY).

FSC 8140

AUTHORIZED FOR HIGHWAY SHIPMENTS ONLY

M.S. Seay NWHL 6/15/76
TECHNICAL DIRECTION AGENT (TDA) DATE

J.E. Kelly 6/15/76
AIRSYSCOM BY DIRECTION DATE

ORIGINATOR Charles McBride 6/15/76 DATE
NAVAL WEAPONS HANDLING LABORATORY
WPNSTA EARLE, NEW JERSEY

Source: https://assist.dla.mil -- Downloaded: 202610-23T23:44Z
Check the source to verify that this is the current version before use.

GENERAL NOTES

1. THIS DOCUMENT PROVIDES DETAILED INSTRUCTIONS FOR TRUCKLOADING THE PRACTICE BOMB MK 76 MODS 4 & 6 WHEN THE BOMBS ARE PALLETIZED IN ACCORDANCE WITH FLEET ISSUE UNIT LOAD WR-54/217.
2. THE PROCEDURES AND PRACTICES DESCRIBED HEREIN ARE INTENDED FOR 40, 42, 44 AND 45 FT TRAILERS WITH BOTH NAILABLE AND NON-NAILABLE FLOORS. TRAILERS SHALL HAVE THEIR AXLES LOCATED IN THE "WESTERN" POSITION (AT THE EXTREME REAR OF THE TRAILER).
3. THE REAR BLOCKING ASSEMBLIES, DETAILS B OR C, SHALL FILL THE VOID BETWEEN THE LADING AND THE TRAILER DOORS WHEN THE DOORS ARE CLOSED.

WARNING

DO NOT USE TRAILERS WITH ROLL-UP DOORS WHEN BLOCKING BY THIS METHOD.

4. SIDE BLOCKING ASSEMBLIES ARE USED WHEN THE TRAILER HAS NON-NAILABLE FLOOR. IF THE TRAILER HAS A NAILABLE FLOOR 2 X 4 DOUBLED SLEEPERS MAY BE SUBSTITUTED FOR THE ASSEMBLIES. INSTALL AS SHOWN ON PAGE 6.
5. THE SIDE BLOCKING ASSEMBLIES, DETAILS A AND F, SHALL FILL THE VOID BETWEEN THE LADING AND THE TRAILER SIDE WALLS.
6. THE SIDE BLOCKING ASSEMBLIES, DETAILS A AND F, ARE IN TWO LENGTHS. ALWAYS USE THE LONGER ASSEMBLY WHEN POSSIBLE USE THE SHORTER ASSEMBLY WHEN REQUIRED BY DIMENSIONAL LIMITATIONS.
7. THE MAXIMUM GROSS WEIGHT OF THE TRACTOR-TRAILER AND THE ALLOWABLE AXLE WEIGHTS ARE THE RESPONSIBILITY OF THE CARRIER. THE CARRIER WILL ADVISE THE SHIPPER OF THESE LIMITATIONS AND THE SHIPPER SHALL LOAD THE TRAILER IN SUCH A MANNER THAT THE TRACTOR-TRAILER WILL NOT EXCEED THESE LIMITATIONS.
8. AFTER BLOCKING AND BRACING HAS BEEN INSPECTED, ATTACH SHIPPING DOCUMENTS TO INSIDE OF TRAILER IN AN ACCESSIBLE AREA, CLOSE AND SEAL TRAILER DOORS.
9. APPLICABLE MATERIAL SPECIFICATIONS: DUNNAGE LUMBER, MM-L-751; NAILS, FF-N-105, TYPE II STYLE 10, COMMON BRIGHT; STRAPPING, QQ-S-781, TYPE I CLASS A.
10. FOR GENERAL TRUCKLOADING PROCEDURES, REFER TO THE GENERAL TRUCKLOADING DOCUMENT, MIL-STD-1320 (NAVY).

FTL AND LTL 40, 42, 44 & 45 FT TRAILERS (VAN)

WARNING

THIS STANDARD PROVIDES FOR INCREASED LOADINGS WHICH PRODUCE HEAVIER GROSS TANDEM AXLE WEIGHTS AND GROSS VEHICLE WEIGHTS. THESE INCREASED WEIGHTS ARE NOT AUTHORIZED IN ALL 50 STATES. BEFORE LOADING MORE THAN 14 UNIT LOADS IN A TRAILER, DETERMINE THAT THE LIMITATIONS OF THE NOTES FOR TABLES I, II, III & IV ARE MET.

1. TABLES I, II, III & IV PROVIDE THE CORRECT LOAD PATTERNS FOR THE LENGTH OF THE TRAILER BEING USED AND THE NUMBER OF UNIT LOADS BEING LOADED.

NOTE

LIGHT WEIGHT TRACTOR-TRAILER - TARE WEIGHT OF TRAILER TANDEM AXLES 6,800 LBS MAXIMUM AND THE DRIVE TANDEM AXLES 10,500 LBS MAXIMUM

STANDARD WEIGHT TRACTOR-TRAILER - TARE WEIGHT OF THE TRAILER TANDEM AXLES 7,800 LBS MAXIMUM AND THE DRIVE TANDEM AXLES 11,500 LBS MAXIMUM

THE ABOVE MAXIMUM WEIGHT MAY BE ADJUSTED UPWARD WHEN EXPERIENCE INDICATES THAT THE WEIGHT RESTRICTIONS CAN BE MET.

2. TO DETERMINE THE CORRECT LOAD PATTERN TO USE, SELECT THE CORRECT TABLE FOR THE LENGTH OF THE TRAILER BEING LOADED. OPPOSITE THE NUMBER OF UNIT LOADS BEING LOADED, FIND THE CORRECT LOAD PATTERN. MAKE CERTAIN THAT THE QUANTITY BEING SHIPPED IS COMPATIBLE WITH THE WEIGHT OF THE TRAILER BEING LOADED AND THE WEIGHT LAWS OF THE STATES THROUGH WHICH THE SHIPMENT IS ROUTED. BLOCK AND BRACE USING THE PRINCIPLES OF THE TRUCKLOADS SHOWN ON PAGES 5 AND 6.
3. THE TYPE OF REAR BLOCKING TO BE USED AT THE REAR OF THE LOAD IS DETERMINED BY THE DISTANCE BETWEEN TRAILER DOORS, WHEN CLOSED, AND THE REAR OF THE LADING. WHEN THIS DISTANCE IS:

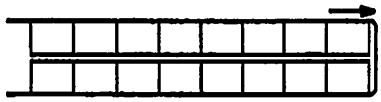
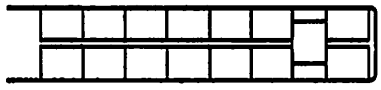
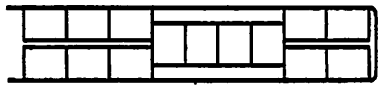
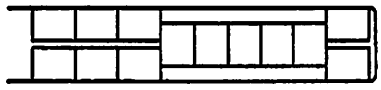
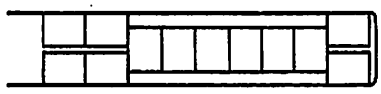
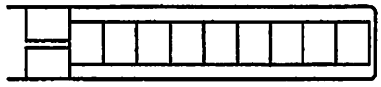
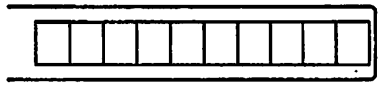
   LESS THAN 12 INCHES - USE FILLER ASSEMBLY, DETAIL C, BETWEEN LADING AND DOORS.
   12 TO 36 INCHES - USE FILLER ASSEMBLY, DETAIL B, AS SHOWN ON PAGE 5.
   GREATER THAN 36 INCHES - USE REAR BLOCKING ASSEMBLY, DETAIL H, AS SHOWN ON PAGE 6. DO NOT USE TRAILERS WITH NON-NAILABLE FLOORS.

4. THE TRUCKLOADS SHOWN ON PAGES 5 AND 6 DO NOT LIMIT SHIPMENT TO THE QUANTITIES SHOWN. THEY ARE PRESENTED TO SHOW THE BASIC PRINCIPLES OF LOADING ONLY.
5. BLOCKING TO PREVENT LATERAL MOTION IS NOT REQUIRED WHEN THE UNIT LOADS ARE ARRANGED IN A DOUBLE ROW PATTERN.

Source: https://assist.dla.mil -- Downloaded: 202610-23T23:44Z
Check the source to verify that this is the current version before use.

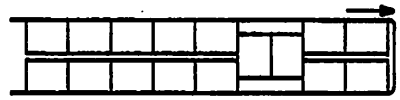
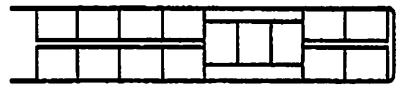
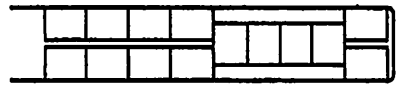
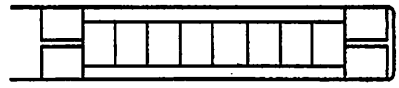
Source: https://assist.dla.mil -- Downloaded: 20261O-237T23:44Z
Check the source to verify that this is the currentversion before use.

TABLE I
40 FT TRAILER

NO. OF UNIT LOADS	LOAD PATTERN (FWD)
16 **	
15 *	
14	
13	
12	
11	
10	

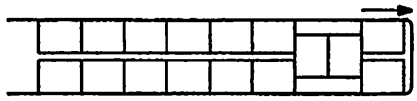
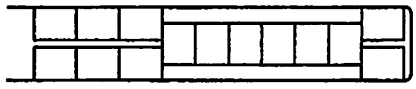
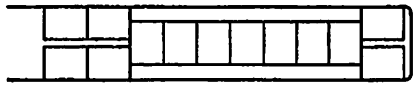
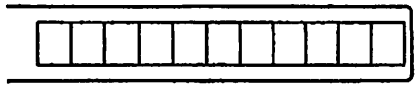
9 OR LESS: USE SINGLE ROW PATTERN.

TABLE II
42 FT TRAILER

NO. OF UNIT LOADS	LOAD PATTERN (FWD)
16 **	
15 *	
14	
13	
12	
11	
10	

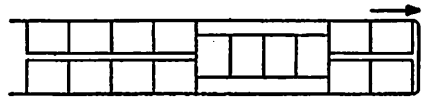
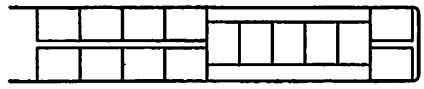
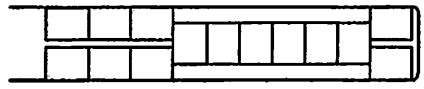
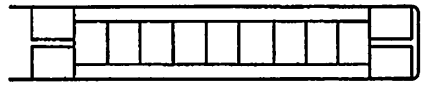
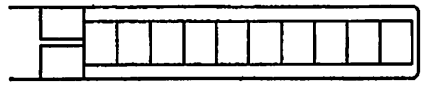
9 OR LESS: USE SINGLE ROW PATTERN.

TABLE III
44 FT TRAILER

NO. OF UNIT LOADS	LOAD PATTERN (FWD)
16 **	
15 *	
14	
13	
12	
11	

10 OR LESS: USE SINGLE ROW PATTERN.

TABLE IV
45 FT TRAILER

NO. OF UNIT LOADS	LOAD PATTERN (FWD)
16 **	
15 *	
14	
13	
12	
11	

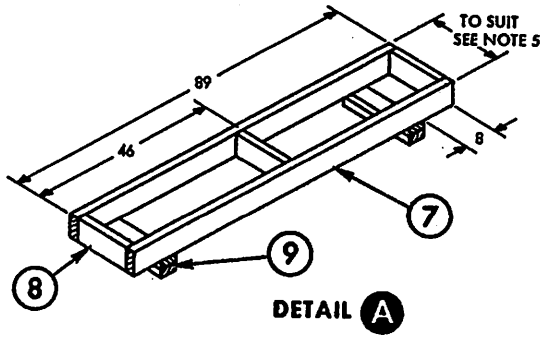
10 OR LESS: USE SINGLE ROW PATTERN.

** AUTHORIZED WHEN ROUTING IS THROUGH STATES PERMITTING 34,000 LBS ON TANDEM AXLES AND A GROSS VEHICLE WEIGHT OF 80,000 LBS. BOTH STANDARD AND LIGHTWEIGHT TRACTOR-TRAILERS MAY BE USED.

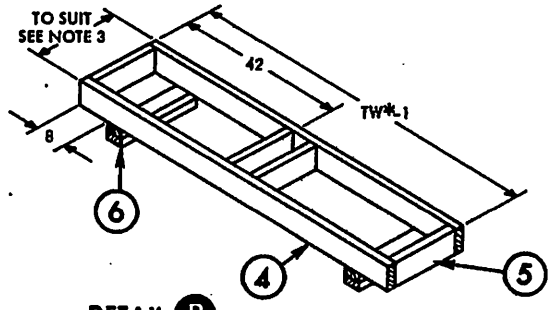
* AUTHORIZED FOR LIGHTWEIGHT TRACTOR-TRAILERS ONLY WHEN ROUTING IS THROUGH STATES LIMITING TANDEM AXLES TO 32,000 LBS AND GROSS VEHICLE WEIGHTS TO 73,280 LBS. AUTHORIZED FOR BOTH STANDARD AND LIGHTWEIGHT TRACTOR-TRAILERS WHEN ROUTING IS THROUGH STATES PERMITTING 34,000 LBS ON TANDEM AXLES AND GROSS VEHICLES WEIGHTS OF 80,000 LBS.

NOTE: WHEN SHIPPING 14 OR LESS UNIT LOADS BOTH STANDARD AND LIGHTWEIGHT TRACTOR-TRAILERS MAY BE USED IN ALL 48 STATES WITHOUT EXCEEDING THE WEIGHT LAWS.

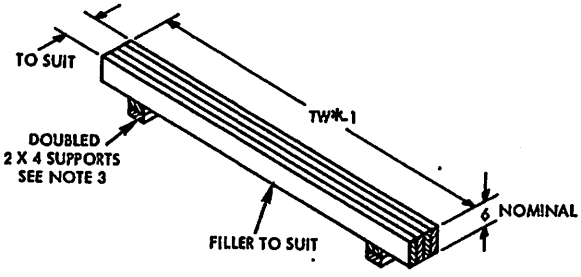
Source: https://assist.dla.mil -- Downloaded: 20264-237T23:44Z
Check the source to verify that this is the current version before use.



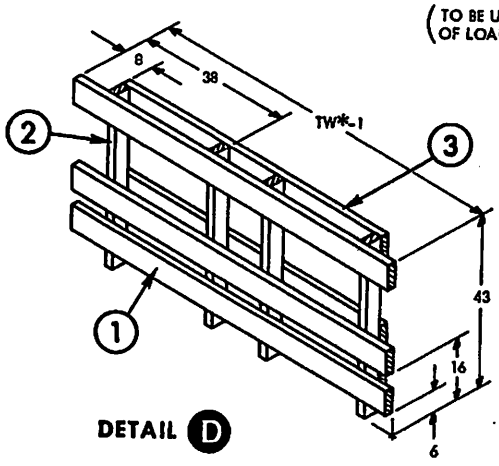
DETAIL A
SIDE BLOCKING ASSEMBLY



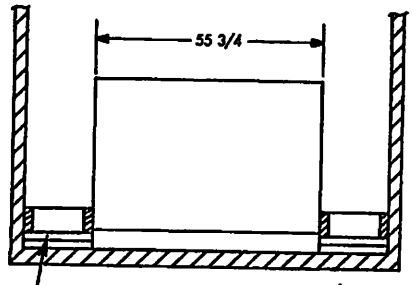
DETAIL B
REAR BLOCKING ASSEMBLY
1 REQUIRED



DETAIL C
SOLID FILL
(TO BE USED WHEN SPACE AT REAR OF LOAD IS LESS THAN 12 INCHES)



DETAIL D
FRONT BULKHEAD
1 REQUIRED

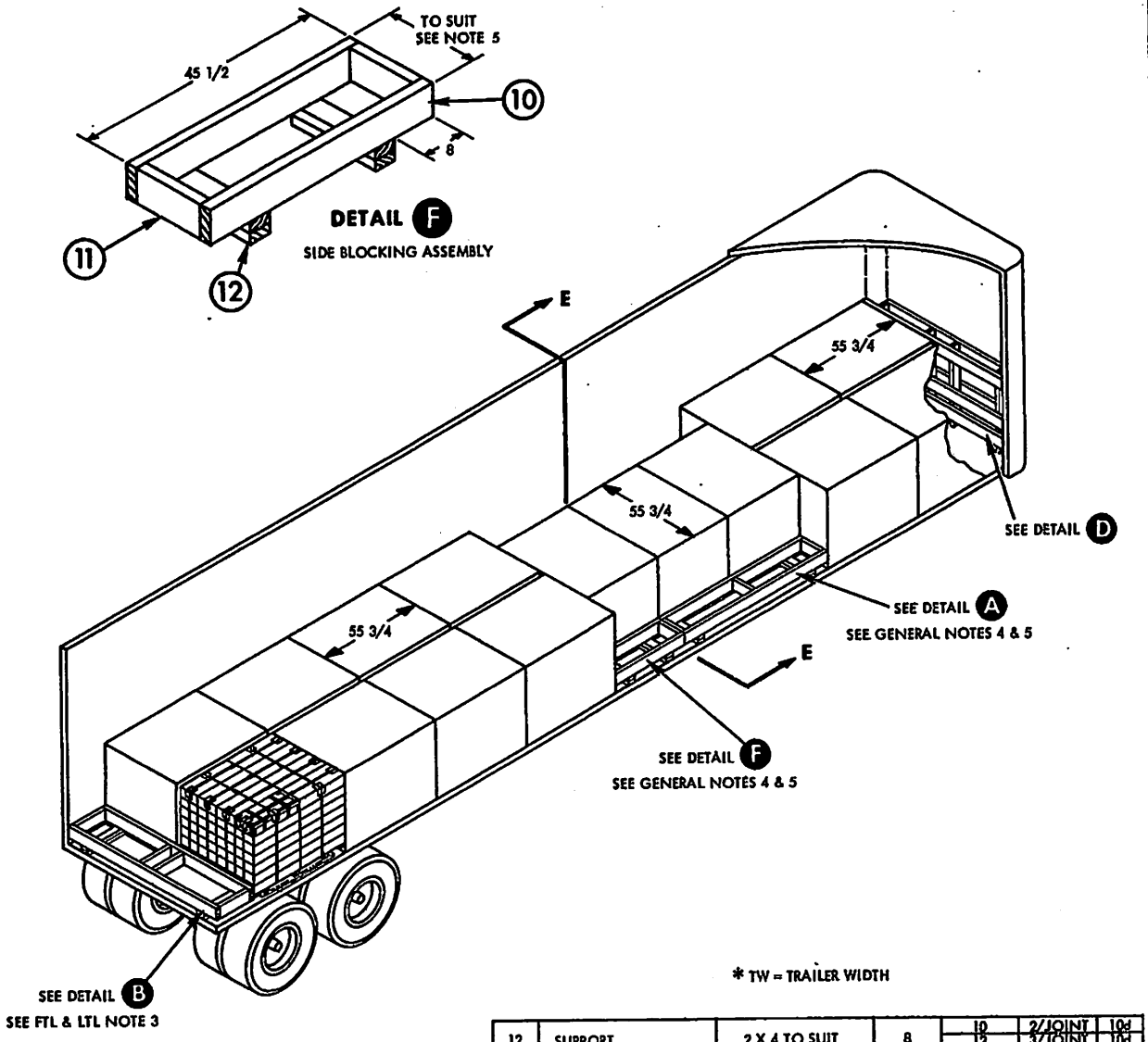


SEE DETAIL A
SEE NOTE 4

SECTION E-E

* TW=TRAILER WIDTH

Source: https://assist.dla.mil -- Downloaded: 20260-23T23:44Z
Check the source to verify that this is the current version before use.



* TW = TRAILER WIDTH

TABLE V

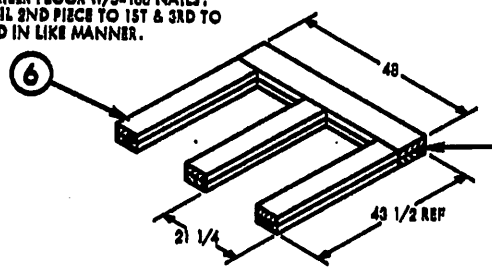
TRUCKLOAD DATA

NUMBER OF UNIT LOADS	TRUCKLOAD WEIGHT LBS (APPROX)
16	48,352
15	45,330
14	42,308
13	39,286
12	36,264
11	33,242
10	30,220

PIECE NO.	DESCRIPTION	SIZE	NO. OF PIECES REQ'D	NAIL TO	NUMBER NAILS	SIZE
LIST OF MATERIALS & NAILING DATA						
12	SUPPORT	2 X 4 TO SUIT	8	~~10~~ 12	~~2/JOINT~~ 3/JOINT	~~10d~~ 16d
11	STRUT	2 X 6 TO SUIT	4	SEE 10&12	-	-
10	LONGITUDINAL	2 X 6 X 45 1/2	4	11	3/JOINT	16d
9	SUPPORT	2 X 4 TO SUIT	8	~~8~~ 9	~~2/JOINT~~ 3/JOINT	~~10d~~ 16d
8	STRUT	2 X 6 TO SUIT	6	SEE 7&9	-	-
7	LONGITUDINAL	2 X 6 X 89	4	8	3/JOINT	16d
6	SUPPORT	2 X 4 TO SUIT	4	~~4~~ 6	~~2/JOINT~~ 3/JOINT	~~10d~~ 16d
5	STRUT	2 X 6 TO SUIT	4	SEE 4&6	-	-
4	CROSSMEMBER	2 X 6 X TW*-1	2	5	3/JOINT	16d
3	FWD CROSSMEMBER	2 X 6 TO SUIT	3	2	3/JOINT	12d
2	VERTICAL	2 X 4 X 43	4	SEE 1 & 3	-	-
1	AFT CROSSMEMBER	2 X 6 X TW*-1	3	2	3/JOINT	12d

TYPICAL LOAD

2 X 6 DOUBLED. NAIL 1ST PIECE TO TRAILER FLOOR W/3-16d NAILS. NAIL 2ND PIECE TO 1ST & 3RD TO 2ND IN LIKE MANNER.



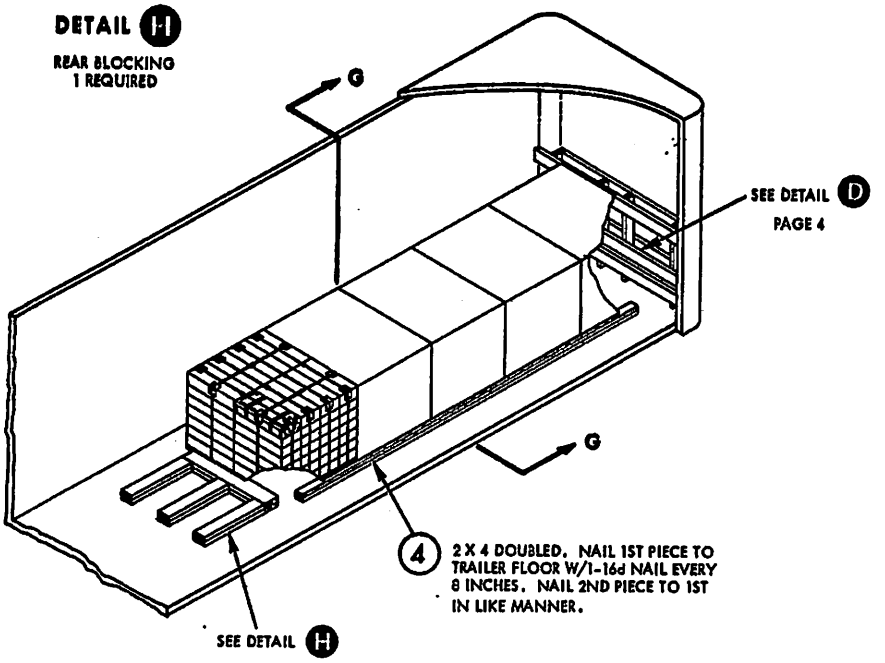
WARNING

DO NOT USE TRAILERS WITH NON-NAILABLE FLOORS.

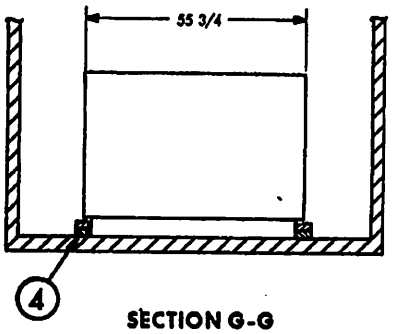
5 2 X 8 DOUBLED. NAIL 1ST PIECE TO TRAILER FLOOR W/6-16d NAILS. NAIL 2ND PIECE TO 1ST & 3RD TO 2ND IN LIKE MANNER.

DETAIL H

REAR BLOCKING
1 REQUIRED



4 2 X 4 DOUBLED. NAIL 1ST PIECE TO TRAILER FLOOR W/1-16d NAIL EVERY 8 INCHES. NAIL 2ND PIECE TO 1ST IN LIKE MANNER.



SECTION G-G

* TW=TRAILER WIDTH

PIECE NO.	DESCRIPTION	SIZE	NO. OF PIECES REQ'D	NAIL TO	NAILS: NUMBER	NAILS: SIZE
6	BACKUP	2 X 6 X 36	9	SEE DETAIL F		
5	CROSSMEMBER	2 X 8 X 48	2	SEE DETAIL F		
4	SLEEPER	2 X 4 X 74 FT	RANDOM LENGTHS	SEE FIELD NOTE		
3	FWD CROSSMEMBER	2 X 6 TO SUIT	3	2	3/JOINT	12d
2	VERTICAL	2 X 4 X 43	4	SEE 1&3	-	-
1	AFT CROSSMEMBER	2 X 6 X TW*-1	3	2	3/JOINT	12d

LIST OF MATERIALS & NAILING DATA

TYPICAL LTL

REVIEW ACTIVITIES:
NAVY-OS, AS

PREPARING ACTIVITY
NAVY-OS
(PROJECT NO. 8140-N338)

Source: https://assist.dla.mil -- Downloaded: 202610-23T23:44Z
Check the source to verify that this is the current version before use.